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Faculty of Engineering (FE)
Department of Electrical and Electronic Engineering (EEE)

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Faculty Name:	Debraj Das		

Capstone Project Title:	Arduino-Based Flood Monitoring and Alert System
Project Group #:	06

SI #	Student Name	Student ID #	Program	Signature
1.	Tajmun Nahar Tisha	23-54386-3	BSc CSE	Tajmun
2.	Zarin Anan	23-55874-3	BSc CSE	Zarin
3.	Lubna Islam Shuzona	24-56570-1	BSc CSE	Shuzona
4.	Kazi Ayesha Afia Rayeesa	23-52794-2	BSc CSE	Ayesha
5.	Saharia Sanzida Liya	23-53055-3	BSc CSE	Liya

Assessment Materials and Marks Allocation:

CO/ CLO Number	CO/CLO Statement	K	P	A	Assessed Program Outcome Indicator	BNQF Indicator	Teaching- Learning Strategy	Assessment Strategy
3	Demonstrate a course project using microcontrollers, sensors, actuators, switches, display devices, etc. that can solve a complex engineering problem in the electrical and electronic engineering discipline through appropriate research.	K8	P1 P3 P7		P.d.1.P3	FS.3	Discussion	Project Report (Literature Review)
4	Explain the complex engineering activities of a course project solving a complex engineering problem of the electrical and electronic engineering discipline through an effective presentation.			A1 A2	P.j.3.A4	SS.2	Discussion	Project Presentation

Assessment Rubrics:

COs	Excellent to Proficient [31-45]	Good [21-30]	Acceptable [11-20]	Unacceptable [1-10]	No Response [0]	Secured Marks
CO3 P.d.1.P3	The outcome of the project demonstrates a course project utilizing microcontrollers, sensors, actuators, switches, display devices, and more, which can address a complex engineering problem in the electrical and electronic engineering field through appropriate research.	The outcome of the project demonstrates a course project utilizing microcontrollers, sensors, actuators, switches, display devices, etc., and also addresses a complex engineering problem in the electrical and electronic engineering discipline through research.	The outcome of the project demonstrates a course project using microcontrollers, sensors, actuators, switches, display devices, etc. but cannot solve a complex engineering problem properly in the electrical and electronic engineering discipline through appropriate research.	The outcome of the project does not demonstrate a course project using microcontrollers, sensors, actuators, switches, display devices, etc. It also could not solve a complex engineering problem in the electrical and electronic engineering discipline through appropriate research.	No Response at all/copied from others/ identical submissions with gross errors/image file printed	
Comments					Total Marks (45)	

Arduino-Based Flood Monitoring and Alert System

Tajmun Nahar Tisha
*Computer Science and
Engineering*
*American International
University-Bangladesh*
Dhaka, Bangladesh
23-54386-3@student.aiub.edu

Lubna Islam Shuzona
*Computer Science and
Engineering*
*American International
University-Bangladesh*
Dhaka, Bangladesh
24-56570-1@student.aiub.edu

Saharia Sanzida Liya
*Computer Science and
Engineering*
*American International
University-Bangladesh*
Dhaka, Bangladesh
23-53055-3@student.aiub.edu

Zarin Anan
Computer Science and Engineering
American International University-Bangladesh
Dhaka, Bangladesh
23-55874-3@student.aiub.edu

Kazi Ayesha Afia Rayeesa
Computer Science and Engineering
American International University-Bangladesh
Dhaka, Bangladesh
23-52794-2@student.aiub.edu

Abstract—Flooding can cause significant damage to human life, property, roads, agricultural areas, and local infrastructure, particularly in regions that experience heavy rainfall and inadequate drainage systems. The early warnings of rain and very quick rises of water levels can enable residents, emergency responders and relevant authorities to take fast preventive measures to minimize potential losses. This project is aimed at developing a flood monitoring and early warning system using an Arduino UNO that integrates a rain sensor and an ultrasonic water level measuring sensor to detect rain and the water level in real-time. The rain sensor can detect the existence of rain and the amount of rainfall; the ultrasonic sensor can measure the distance between the water surface and the sensor, so that the water level changes can be measured. The information gathered by the sensors is fed into the Arduino UNO, which then analyzes the data to determine the environmental state as 'Normal', 'Warning' or 'Danger'. Current condition is printed on LCD and thus users get the information of the monitored area clearly and instantly. A buzzer and

LED indicators give a local warning during dangerous conditions to bring attention to the danger. Furthermore, a GSM module sends SMS alerts to residents and local authorities in case of a critical water level detection. It is a continuous monitoring process that provides a simple, affordable and practical approach for flood-risk awareness and early warning in flood vulnerable communities and facilitate prompt flood emergency decision making process effectively.

Keywords— *Arduino UNO, flood monitoring, rain sensor, ultrasonic sensor, water level, LCD, GSM, buzzer, LED, early warning.*

I. INTRODUCTION

2.1 Background of Study and Motivation

Floods are one of the most frequent natural disasters, which can come with devastating effects on human life, properties, roads, farms and local infrastructure.

It can cause high localized rainfall, which can lead to dangerous conditions in many areas for nearby communities. It is therefore important to monitor the rainfall as well as changes in the water-levels real-time, so that potential flood conditions can be detected and warnings given in due time. Existing flood monitoring systems can involve significant acquisition costs, installation complexity, manual check frequency or monitoring centers. Such systems are subject to these requirements that can be challenging in remote, rural and small community settings, and where resources are limited. The new era of low cost micro-controllers and sensors means making a more basic monitoring system, and gathering environmental data on a continual basis and alerting immediately.

The aim of this project is to build a low-cost, easy-to-understand prototype for flood monitoring and early warning with the use of Arduino UNO. It is a combination of rain sensor and ultrasonic sensor for rain and water level measurement. Processed information from the collected data is used by the Arduino UNO to determine the current conditions of the monitored area. The system gives a warning locally on an LCD display, LED and buzzer and when a dangerous condition is detected the warning is sent to predefined recipients using an SMS via GSM module. The project will showcase the capacities of low-cost electronic components to aid in raising awareness in the context of flood risk, in continuous monitoring, and in early warning.

2.2 Project Objectives

The primary aim of this project is to design and develop rain water level monitoring and early warning system using Arduino UNO which can monitor the rain water level continuously. The system is capable of detecting the rainfall with rain sensor and measuring water level with non-contact ultrasonic sensor. This sensor information is sent to the Arduino UNO which determines the environment's current classification and outputs an alert message to either a Normal, Warning or Danger level. The system will also be designed to indicate the current status on an LCD panel and give immediate warning locally through LED light and buzzer. Furthermore, a GSM module is embedded to allow sending SMS alerts to pre-determined

recipients in the case of a critical or dangerous situation. The overall objective for the project is to create a simple, low cost, reliable, and continuous operating prototype to demonstrate the operation of basic functionality of flood detection, local warning and remote notification.

2.3 A Brief Outline of the Report

This report is divided into a series of chapters, in order to present the development of the proposed flood monitoring and early warning system. The background information, motivation, project goals and report structure are described in chapter 1. Literature review is presented in Chapter 2 and an overview of the various types of flood monitoring and warning systems available as well as their technologies, methodologies, benefits, and limitations are discussed. Chapter 3 deals with the proposed system, which contains the system architecture, hardware components, software requirements, circuit connections and overall working process of the proposed system. Chapter 4 discusses the implementation and testing of the system which involves rainfall detection system and the water-level measuring system, flood condition classification, LCD display, LED and buzzer alert system and GSM-based SMS notification system. The results of system testing and overall performance and limitations of the prototype are presented and discussed in Chapter 5. The report ends with Chapter 6, where the project's successes are summarized and possible future enhancements are discussed to improve the reliability, functionality, and applicability of the system for real-world flood monitoring use.

II.LITERATURE REVIEW

Disasters caused by flooding are still a significant threat to human lives, property, infrastructure and economy worldwide, and there is a growing interest in automated and technologically driven Early Warning Systems (EWS) research. Microcontroller, IoT, WSN, and ES applications in order to deliver local information and response time to vulnerable communities have been explored recently. The goal

with these systems is to enhance flood monitoring through continual monitoring of environmental data and actives that notify when potentially hazardous conditions are detected.

Ganancias et al. [1] developed a microcontroller-based EWS for remote, high-risk regions, adopting a multi-hazard approach that monitors both flood levels and landslide risks through water-level, soil-moisture, and ground-displacement sensors. The study used a descriptive-evaluative design and integrated environmental sensors with a GSM module to support offline functionality in areas with unstable internet connectivity. The system achieved high compliance with ISO 25010 quality standards. However, institutional response remained largely reactive due to incomplete geographic coverage and limited equipment availability.

Ayorinde et al. [2] focused on community-level implementation in Kroo Bay, Sierra Leone, using an Arduino Uno with an HC-SR04 ultrasonic sensor and a SIM800L GSM module for SMS alerts. The system achieved a measured accuracy of ± 1.8 cm and an SMS latency of 5–10 seconds. Unlike purely technical studies, this work also addressed physical security by redesigning the hardware casing so that it could be cemented into walls after early units were stolen. However, GSM network failure during extreme rainfall and the lack of long-term dry-season testing remained important limitations.

Ramadan and Sulistya [3] proposed a low-cost IoT solution using a conductive K-0135 water-level sensor instead of a non-contact ultrasonic sensor. Their approach aimed to reduce false alarms through threshold detection under both static and dynamic rainfall conditions. The system used a NodeMCU ESP8266 and HTTP-based communication and achieved a 100% communication success rate in laboratory testing. However, the system was not validated under real-world conditions involving factors such as debris accumulation, environmental interference, or long-term sensor degradation.

Faturochman et al. [4] developed an ESP32-based monitoring system integrated with a Telegram chatbot to provide real-time public alerts. The system followed the waterfall development method and used sensor measurements to determine water-level conditions. Across seven experiments, the sensor-calculated water levels matched the manual

measurements. However, the system relied primarily on simple threshold-based logic and did not integrate atmospheric or rainfall-related data, limiting its ability to provide a broader assessment of flood conditions.

Susanti et al. [5] applied ultrasonic sensing specifically to tidal flood monitoring in Semarang, Indonesia, using an ESP32-S2 microcontroller and a high-precision A01NYUB ultrasonic sensor. The system achieved a tidal-level measurement accuracy with an RMSE of 0.08943. The study also found that physical distancing of approximately 166 cm from the monitoring tower was required to prevent the sensor from detecting wave noise instead of the mean sea level. This finding highlights the importance of appropriate sensor positioning and digital signal filtering, particularly in coastal environments where wave motion can affect ultrasonic measurements.

Mansuit et al. [6] expanded the sensing scope by integrating water-level measurement with atmospheric parameters, including temperature, humidity, and barometric pressure. The system used AHT20 and BMP280 sensors along with an adaptive sampling technique and a Django-based web application. The proposed approach provided a scalable, cloud-based framework for environmental monitoring. However, the system was evaluated using simulated data rather than complete real-world hardware deployment, leaving its practical performance under actual environmental conditions unverified.

III. METHODOLOGY AND MODELING

4.1 INTRODUCTION

Floods are one of the most common natural hazards and can cause significant damage to human lives, properties, roads, and local infrastructure. Early detection of rainfall and rising water levels can help people and authorities take appropriate action before the situation becomes critical. The proposed project develops a low-cost Arduino UNO-based flood monitoring and early warning system that continuously monitors water level and rainfall conditions. The system uses an HC-SR04 ultrasonic sensor to measure water level and a rain sensor to detect rainfall. The collected information is processed

by the Arduino UNO and classified into different conditions based on predefined threshold values. An LCD displays the current condition, while LEDs and a buzzer provide local visual and audio warnings. A GSM module is also included to send SMS alerts to predefined recipients during dangerous conditions. The system is first tested through simulation and then evaluated through experimental testing to verify its functionality and response.

4.2 Working Principle of the Proposed Project

The proposed flood monitoring system works by continuously collecting environmental information from the ultrasonic and rain sensors. The HC-SR04 ultrasonic sensor is positioned above the water surface and measures the distance between the sensor and the water. As the water level rises, the measured distance decreases. The Arduino UNO processes this distance measurement and converts it into a water-level value. At the same time, the rain sensor detects the presence of rainfall and provides supplementary information about the environmental condition.

The Arduino compares the measured water level with two predefined threshold values, $T_{warning}$ and T_{danger} . When the water level is below $T_{warning}$, the system considers the condition Safe and continues normal monitoring. When the water level reaches $T_{warning}$ but remains below T_{danger} , the system enters the Medium or Warning condition. When the water level reaches or exceeds T_{danger} , the system identifies the condition as Danger. In the Danger state, the buzzer and appropriate LED indicator are activated to provide an immediate local warning. The GSM module is then used to send an SMS alert to predefined recipients. The current condition is continuously displayed on the LCD. After completing one monitoring cycle, the Arduino waits for a fixed sampling interval and repeats the process continuously.

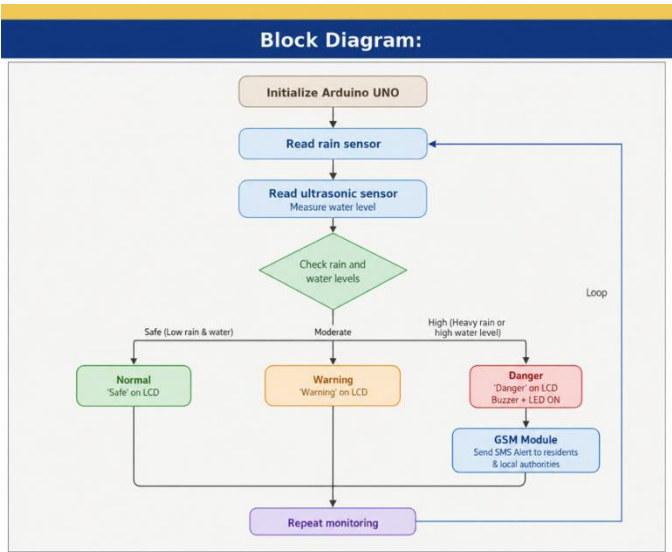


Fig. 1. System block diagram: Initialize → Read Water Level → Check Water Level → {Normal / Warning / Danger} → Repeat Monitoring.

4.2.1 Process of Work (Experimental and Simulation Processes)

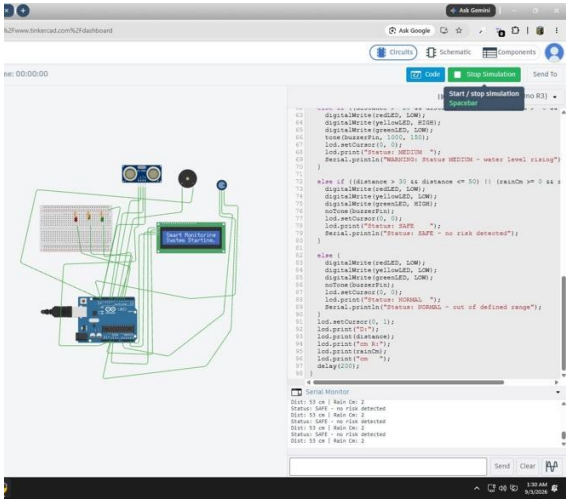


Fig. 2. Initial Setup of the Flood Monitoring System

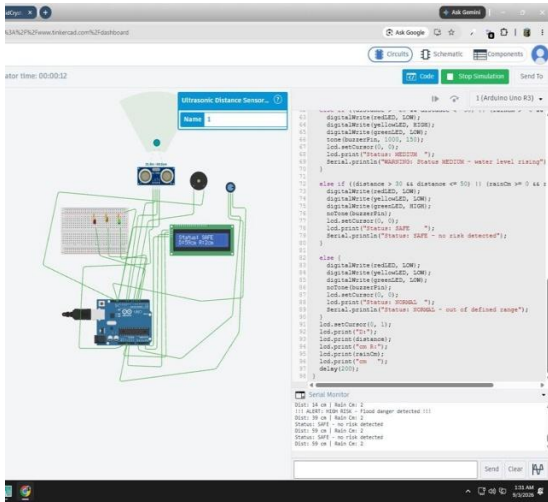


Fig. 3 Simulation Result under Safe Condition

Fig.2 The initial simulation setup consists of an Arduino UNO, ultrasonic distance sensor, rain sensor, LCD display, LEDs, and buzzer. The Arduino UNO acts as the main controller and receives input from the sensors. The LCD is used to display the current monitoring status, while the LEDs and buzzer provide visual and audible indications during abnormal conditions. Fig.3 using the safe condition, the measured distance from the ultrasonic sensor remains within the predefined safe range, while the rain sensor does not indicate a critical rainfall condition. The LCD displays “**Status: SAFE**”, and the green LED remains active. The buzzer remains OFF, indicating that no immediate flood risk has been detected.

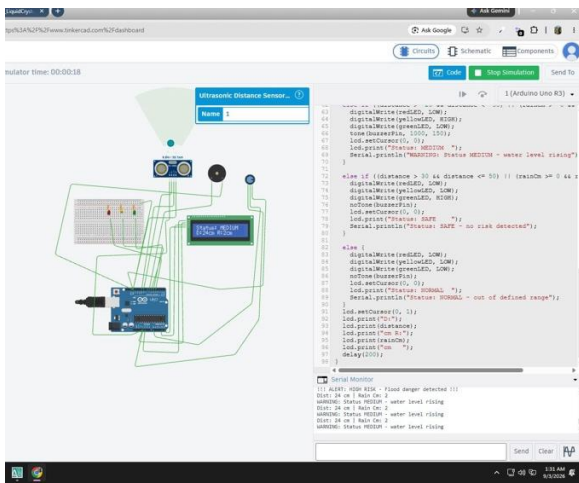


Fig. 4. Simulation Result under Medium Condition



Fig. 5. Simulation Result under Danger Condition

In Fig.4. the medium condition, the ultrasonic sensor detects a reduced distance, indicating that the water level is rising. The system identifies this condition as a warning state. The LCD displays “**Status: MEDIUM**”, and the yellow LED is activated. The buzzer provides an alert to indicate that the water level requires monitoring and may become critical if it continues to rise. Fig.5. During the danger condition, the ultrasonic sensor detects a significantly reduced distance, indicating a high water level. The system identifies the situation as a high-risk flood condition. The LCD displays “**Status: HIGH RISK**” or the corresponding danger message, while the red LED and buzzer are activated. The Serial Monitor also displays a high-risk flood alert, confirming that the system has successfully detected the dangerous condition.

4.3 Description of the Components

The proposed system is comprised of a number of hardware components that can be combined to carry out the monitoring and warning functions in the flooding context. Arduino UNO is the main microcontroller that receives the input from sensors, process the received data, implements the classification logic depending on the threshold value and drives the output devices. The water-level

measurement is performed via HC-SR04 ultrasonic sensor, which is a non-contact type water-level sensor. It is a device that emits an ultrasonic pulse toward the water's surface, and works by computing the distance based on the echo reflection. The rain sensor is utilized for rainfall conditions detection, and for the monitoring process, it will give additional environmental information.

A 16×2 LCD is utilized to depict the present status of the system, and also to present the current information to the user. The different system conditions are indicated by LEDs, and a buzzer gives a prompt audible warning of a dangerous condition. The alarm is sent via SMS to set up predefined cell phone numbers when the water level is reached by the GSM/GPRS module, which can be either SIM800L or any compatible module. The Arduino and peripherals are fed power as needed by a regulated power supply. These parts, when combined, can create an early warning system that can identify environmental changes and alert both locally and remotely with minimal cost.

4.4 Test/Experimental Setup

The testing and experimental setup was designed to evaluate the functionality and response of the proposed flood monitoring system under different environmental conditions. During the simulation stage, the circuit was developed in Tinkercad and connected according to the proposed system architecture. The ultrasonic sensor, rain sensor, LCD, LEDs, and buzzer were tested using different input conditions. The water-level values were varied to represent Safe, Medium, and Danger conditions, and the corresponding system responses were observed. The LCD status, LED indicators, and buzzer activation were recorded for each test condition.

For the physical experiment, the Arduino UNO and other hardware components were assembled on a breadboard. The HC-SR04 sensor was placed at a fixed position above a water container, allowing the water level to be increased gradually. Different water levels were created to test the predefined thresholds. The rain sensor was also exposed to different levels of moisture to verify its response to rainfall. For each test, the sensor readings and system outputs were observed. The LCD was checked to confirm the

correct status display, while the LEDs and buzzer were monitored for appropriate warning signals. When the Danger threshold was reached, the GSM module was tested to verify SMS transmission to the predefined recipient. The results from both simulation and experimental testing were compared with the expected system behavior to determine whether the proposed system operated correctly and consistently.

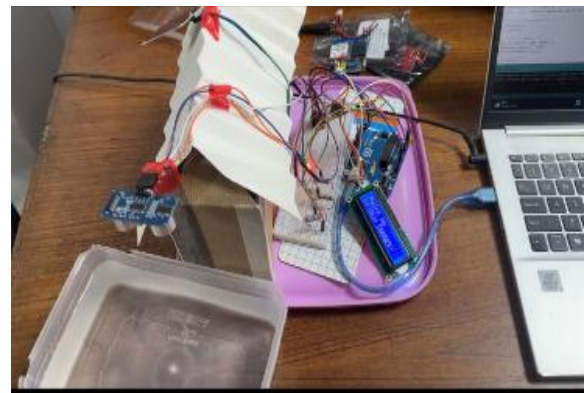
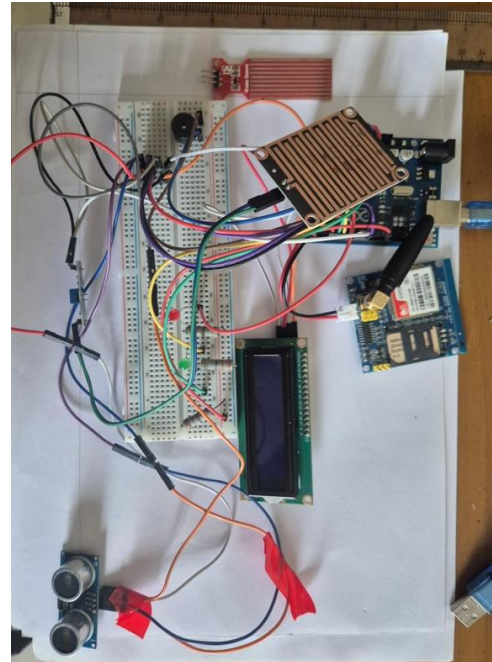


Fig. 6. Assembled hardware prototype of the flood monitoring and alert unit.

5.1 Simulation/Numerical Analysis

The proposed flood monitoring and alert system was first evaluated through Tinkercad simulation to verify the functionality of the Arduino UNO, HC-SR04 ultrasonic sensor, rain sensor, LCD, LEDs, and

buzzer. The simulation was performed under different water-level and rainfall conditions to observe how the system responds to changes in the monitored environment. The ultrasonic sensor provided distance measurements representing the water level, while the rain sensor provided information about rainfall conditions. These sensor inputs were processed by the Arduino UNO and compared with predefined threshold values.

Based on the programmed threshold logic, the system classified the monitored condition into different states, including Setup, Safe, Medium, and Danger. In the Safe condition, the measured distance remained within the predefined safe range, resulting in the activation of the green LED and the display of "Status: SAFE" on the LCD. When the water level increased and the measured distance decreased, the system entered the Medium condition. In this state, the yellow LED was activated and the LCD displayed "Status: MEDIUM". When the water level reached the dangerous threshold, the system classified the condition as Danger or High Risk. The red LED and buzzer were activated, while the LCD and Serial Monitor displayed the corresponding warning message. These simulation observations confirmed that the programmed threshold-based logic responded appropriately to different input conditions.

5.2 Measured Response/Experimental Results

After completing the simulation, the physical prototype was assembled to evaluate the practical response of the proposed system. The Arduino UNO was connected to the HC-SR04 ultrasonic sensor, rain sensor, LCD, LEDs, and buzzer according to the designed circuit. The ultrasonic sensor was positioned above a water container, and the water level was gradually changed to observe the response of the system. The rain sensor was also tested under different moisture conditions to verify its ability to detect rainfall.

During the experimental testing, the system successfully detected changes in the water level and produced different output conditions according to the programmed thresholds. When the water level remained low, the system indicated the Safe condition through the LCD and green LED. As the

water level increased, the system changed to the Medium condition and activated the yellow LED to indicate that the water level required attention. When the water level reached the defined Danger threshold, the red LED and buzzer were activated to provide an immediate warning. The LCD displayed the corresponding danger status. The GSM module was intended to provide an additional remote SMS alert during the Danger condition. The experimental observations were recorded and used to evaluate whether the physical prototype produced the expected responses.

5.3 Comparison between Numerical and Experimental Results

The proposed system is comprised of a number of hardware components that can be combined to carry out the monitoring and warning functions in the flooding context. Arduino UNO is the main microcontroller that receives the input from sensors, process the received data, implements the classification logic depending on the threshold value and drives the output devices. The water-level measurement is performed via HC-SR04 ultrasonic sensor, which is a non-contact type water-level sensor. It is a device that emits an ultrasonic pulse toward the water's surface, and works by computing the distance based on the echo reflection. The rain sensor is utilized for rainfall conditions detection, and for the monitoring process, it will give additional environmental information.

A 16×2 LCD is utilized to depict the present status of the system, and also to present the current information to the user. The different system conditions are indicated by LEDs, and a buzzer gives a prompt audible warning of a dangerous condition. The alarm is sent via SMS to set up predefined cell phone numbers when the water level is reached by the GSM/GPRS module, which can be either SIM800L or any compatible module. The Arduino and peripherals are fed power as needed by a regulated power supply. These parts, when combined, can create an early warning system that can identify environmental changes and alert both locally and remotely with minimal cost.

5.4 Cost Analysis

A cost analysis was carried out to evaluate the affordability of the proposed flood monitoring and early warning system, since one of the main design objectives was to develop a low-cost solution suitable for resource-constrained communities. Table I summarizes the approximate cost of the major hardware components used in the prototype implementation.

Component	Approximate Cost (BDT)
Arduino UNO R3	850
HC-SR04 Ultrasonic Sensor	120
Rain Sensor Module	100
16x2 LCD Display	180
GSM Module (SIM800L)	650
LEDs (Green, Yellow, Red)	15
Buzzer	30
Breadboard and Jumper Wires	250
Power Supply/Battery	200
Total	2395

TABLE I. APPROXIMATE COMPONENT COST BREAKDOWN

The total estimated hardware cost of the prototype is approximately 2395 BDT (around \$20 USD), which is significantly lower than existing commercial flood monitoring systems that often require centralized infrastructure or specialized industrial-grade sensors costing several times more. This low overall cost supports the project's objective of providing an affordable, replicable, and easily deployable model for flood-prone rural and semi-urban communities, particularly since survey respondents identified cost as one of the primary barriers to adopting existing flood-warning solutions.

5.5 Limitations in the Project

Although it has fulfilled the main objectives, a number of limitations, which should be taken into account, have been identified in the proposed system. GSM based SMS alert functionality was

partially verified, although the module was successfully integrated and configured, it was not possible to prove the end-to-end SMS delivery to predefined recipient under all testing conditions because of network signal availability during experiments.

Second, some environmental factors that could influence HC-SR04 ultrasonic sensor accuracy were not considered in this study, e.g., debris accumulating on the sensor, floating objects, and turbulence on the water surface. Thirdly, the system uses a set of fixed threshold values $T_{warning}$ and T_{danger} , which were determined for a certain test container and could be subject to re-calibration in other water bodies or in different seasons.

Fourthly, the prototype was tested for a limited period of time, and this is not indicative of the long term reliability and durability of the sensor when continuously deployed outdoors, in rain, humidity and temperature extremes. Finally, the current design did not include a weatherproof enclosure and a battery backup, which constrains the ability for continuous, unattended use in the field. The system could be made more robust and more applicable to the real world by overcoming these limitations as described in the Future Endeavors section.

V.CONCLUSION AND FUTURE ENDEAVORS

This report presented the design and simulation-based modeling of a low-cost Smart Flood Monitoring and Early Warning System built on an Arduino Uno, an ultrasonic water-level sensor, a rainfall sensor, and a GSM module. Guided by a 56-respondent community survey — in which 75% of respondents had experienced flooding and identified accurate water-level detection and early alerts as top priorities — the system was designed around a threshold-based classification pipeline that triggers local (LCD, buzzer, LED) and remote (SMS) alerts when water levels cross predefined danger thresholds. The design emphasizes affordability and ease of maintenance to address the cost and maintenance barriers most frequently cited by survey respondents.

Once hardware testing is complete, the system is expected to demonstrate reliable, low-latency detection of rising water levels and timely alert dispatch, offering a scalable and replicable model for flood-prone rural and semi-urban areas of Bangladesh. Future work includes extending the threshold-based logic toward a basic predictive algorithm using historical rainfall and water-level trends, integrating solar power for off-grid deployment, adding a companion mobile application for real-time public alerts (as favorably rated by

55.4% of surveyed respondents), and conducting extended field testing across multiple flood seasons to assess long-term reliability.

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